

Business Programming for E&BI: Excel

Christoph Walsh

Introduction

- In this slide deck we will learn some important Excel skills.
- These slides are accompanied by a set of video demonstrations.
- *Note:* this course assumes you already have some basic knowledge of Excel. So we will go through some of the more basic material very quickly.

What is Excel?

- Excel is a *spreadsheet editor*.
- Spreadsheets organize data (numbers and text) in *tabular form*.
 - ▶ These are tables of *cells* organized in *rows* and *columns*.
- In a *table*:
 - ▶ Columns typically represent a variable types (such as the date, sales, expenditures).
 - ▶ Rows typically represent items/records/observations for those variables.
- For example, an ice-cream salesman might record daily sales and the daily temperature in a spreadsheet:

Date	Sales	Temperature
10-08-2023	350	23
11-08-2023	400	26
12-08-2023	380	24

Spreadsheets

- Spreadsheets and Excel are useful for many things:
 - ▶ Organizing and storing many types of data, such as financial information.
 - ▶ Performing statistical analyses on data.
 - ▶ Creating plots and charts of data.

Ancient Spreadsheets

- Spreadsheets are so useful that people have stored data in tabular format for literally thousands of years.



Figure 1: 3700-year-old Babylonian Clay Tablet.

Less-Ancient Spreadsheets

- Until recently, companies recorded sales and costs in accounting ledgers (and some still do).

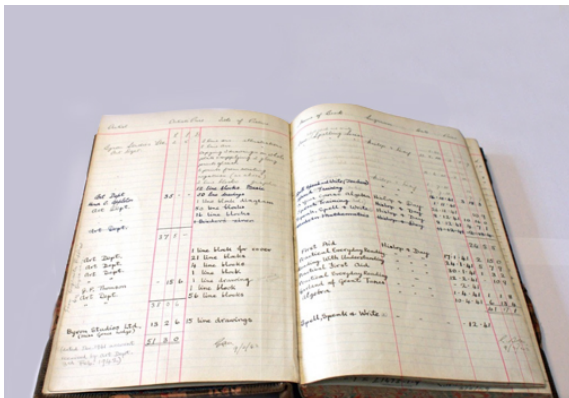


Figure 2: Paper Accounting Ledger.

Digital Spreadsheets

- Spreadsheet software appeared on personal computers as soon they became widespread.



The image shows a screenshot of the VisiCalc spreadsheet application. The window title is 'C11 <L> TOTAL' and the cursor is at cell 'C1' with the value '25'. The spreadsheet has four columns: A (ITEM), B (NO.), C (UNIT), and D (COST). The data is as follows:

	A	B	C	D
1	ITEM	NO.	UNIT	COST
2	MUCK RAKE	43	12.95	556.85
3	BUZZ CUT	15	6.70	101.25
4	TOE TONER	250	49.95	12487.50
5	EYE SNUFF	2	4.95	9.90
6			SUBTOTAL	13155.50
7			9.75% TAX	1282.66
8			TOTAL	14438.16

Figure 3: VisiCalc in 1979.

- MS Excel, LibreOffice Calc and Google Sheets are modern versions of these.

Useful Excel Features

- For the rest of this slide deck we will go over a number of useful Excel features.
- Many Excel features we will talk about will also work in other spreadsheet software, like LibreOffice Calc and Google Sheets.

Cell Formats

- When you input data into Excel it tries to guess the intended format.
 - ▶ For example, if it is a date, a number, a monetary amount, or just text.
 - ▶ If we type in 2/12/23, it guesses that it's a date and formats it to 02/12/2023.
- If we want to change this, we can use the "Format Cells" option.
 - ▶ Select the cell(s) we want to change, right-click, and choose "Format Cells..."

Exercises:

- 1 Type 123.45 in a cell and convert it to a currency (in euros).
- 2 Type in today's date in the format dd/mm/yyyy and format it to US-style: mm/dd/yyyy.

Cells with Long Streams of Text

- Sometimes cells contain very long pieces of text that run over other columns.
- If we want to change this we have a couple of options.
 - ① Use the “Wrap Text” option under Home -> Alignment.
 - ② Insert line breaks using Alt + Enter at the points we want the text to run onto a new line.

Exercises:

- ① Go to <https://loremipsum.io/> and copy the famous (long) Latin filler text (*Lorem ipsum dolor sit amet...*). Add it to a cell and use the wrap text function.
- ② Do the same as (1) but manually insert some line breaks instead.

Excel as a Calculator

- We can use Excel as a basic calculator.
- If we wanted to add $2 + 3$, we can go to an empty cell and type `=2+3` and Enter. The cell would then turn into 5.
 - ▶ In the *Formula bar*, we can still see the original formula `=2+3`.
- To subtract, we use `-`. For example, `=3-2`.
- To multiply, we use `*`. For example, `=2*3`.
- To divide, we use `/`. For example, `=6/2`.
- To take powers, we use `^`. For example, `=3^2`.
- We can also perform multiple operations together using parentheses. For example, if we want to calculate:

$$\frac{12 + 4}{2 \times 4} = \frac{16}{8} = 2$$

We would do `=(12+4)/(2*4)`.

- **Exercise:** Use Excel to calculate $\frac{2 \times 4}{2^2}$.

Cell Referencing

- In Excel, rows are referenced by numbers and columns by letters.
- The cell in the 2nd column and 3rd row can be referenced with B3.
- This referencing will be very useful for calculations, and for using functions.
- We can also refer to multiple cells:
 - ▶ To refer to rows 3-7 in the 4th column, we write D3:D7.
 - ▶ To refer to rows 1-5 in columns 1-2, we write A1:B5.
- We can also refer to cells on other Excel *Sheets*, which are like tabs in the same document.
 - ▶ To refer to cell A1 in the sheet called Sheet2, we write Sheet2!A1.

Calculations in Excel using Cell Values

- We can use cell references to make calculations.

	A	B
1	30	40
2	20	50

- If we want to add A1 and A2 together ($30 + 20$), we go to an empty cell (for example, A3), and type `=A1+A2` and Enter.
 - ▶ We could also type `=` and then click on the cell A1, type `+`, and then click on the cell A2, and then press Enter.

Functions

- We can perform calculations on a cell or group of cells using in-built *functions*.
- If we want to add numbers in a column together, we can use the SUM function.
- For example, let's say we want to add the numbers in column A like in the previous slide:

	A	B
1	30	40
2	20	50

- We can go to a new cell, type =SUM(A1:A2) and Enter.
- Here SUM is the *name* of the function. In Excel these are always with capital letters.
- The term in the parentheses is called the *argument* of the function. In this case it's the cells we want to sum.

Some Common Functions to Summarize Values

In what follows below, *cells* refers to a range of cells, like A1:A2 in the previous slide.

- =AVERAGE(*cells*) – gets the average of *cells*.
- =MEDIAN(*cells*) – gets the median of *cells*.
 - ▶ The median is the value “in the middle” when we sort the numbers.
 - ▶ If there is an even number of numbers it is the average of the two numbers in the middle.
 - ▶ For example, the median of [4, 2, 1, 3] is 2.5, because the average of the two numbers in the middle (2 and 3) is 2.5.
 - ▶ **Exercise:** Populate cells with the numbers [4, 2, 1, 3] and calculate the median.
- =MAX(*cells*) – gets the largest number in *cells*.
- =MIN(*cells*) – gets the smallest number in *cells*.

Adding/Deleting Rows/Columns

- Suppose we have the following table, but we want to add a new row between rows 1 and 2.

	A	B
1	30	40
2	20	50

- We can add 1 new row above row 2 by right-clicking on the “**2**” row index and clicking the Insert option.
 - ▶ What was row 2 then becomes row 3, with an empty row in its place.
- We can also add multiple rows at the same time. For example, to add 4 new rows between rows 1 and 2, we do the following steps:
 - ▶ First select rows 2-5. Click on the “**2**” index for row 2, hold Shift, then click on the “**5**” index for row 5.
 - ▶ Then right click on one of the selected row indices and press Insert. 4 new rows should appear.

Adding/Deleting Rows/Columns

- Adding new columns is very similar. We just right-click on the column index instead. Insert will then add a new column to the left.
- **Exercise:** Add 2 columns between A and B in the previous table.
- To delete a row or column, we select the row/column, right click, and then press the Delete option.

Freezing Panes

- Often we have Excel files with many values that we can't see on our screen all at once.
- If you scroll down to examine values, you can no longer see the first row which often has the names of the variables in the columns.
- We can *freeze* the top row so that we always see the top row when we scroll down.
- To do this, we go to “View” → “Freeze Panes” → “Freeze Top Row”.
- We can also freeze any selection of cells by using the “Freeze Panes” option.

Sorting Ascending or Descending

Suppose we want to sort the data by the values in column A:

	A	B
1	30	40
2	20	50
3	25	45

- We click on the column (by clicking A).
- We go to the “Data” tab and press the $\frac{A}{Z} \downarrow$ button.
- We tick “Expand the selection”. We then get:

	A	B
1	20	50
2	25	45
3	30	40

Sorting Ascending or Descending

- The “Expand the selection” option keeps the cells in B together with their corresponding cell in A, so the cells in B change position as well.
- If we don’t want to do that, we tick “Continue with the current selection”.
- **Exercise:** Sort the data by the values in column B but *descending* (largest to smallest). Also expand the selection.

Transposing

- Sometimes we have data in a row that we want in a column, or vice versa.
- For example, suppose we have:

	A	B
1	30	40
2	20	50
3	25	45

- But we want to convert it to:

	A	B	C
1	30	20	25
2	40	50	45

Transposing

- We need to select and copy the values we want to transpose.
- We then go to an empty part of the spreadsheet.
- We right click, “Paste Special”, and then tick “Transpose”.

Filtering

Suppose we have the following data on the sales of 3 different products in 3 different years:

	A	B	C
1	product	year	sales
2	1	2020	1200
3	1	2021	2400
4	1	2022	1400
5	2	2020	8900
6	2	2021	9700
7	2	2022	7400
8	3	2020	3300
9	3	2021	4400
10	3	2022	2100

If we want to see the sales for all 3 products in 2022 only, we can use the *filter* tool.

Filtering

- Go to the “Data” tab and press the “Filter” toolbar button.
- Little down arrows appear on the column names product, year, and sales.
- To see the rows only for 2022, we click on the down arrow next to year.
- We deselect all rows by unticking “Select all” and then selecting “2022”.
- We then only see the column names and the values for 2022.
- To go back to normal, you can click “Clear” next to the Filter toolbar button.

Filtering

- You can also try more complex filter operations. For example, show all years bigger or equal to 2021 (both 2021 and 2022).
- We go to the small down arrow next to Year, click “Number Filters” → “Greater Than Or Equal To”.
- We then put 2021 into the box (greater than or equal to 2021).
- We then see the data only for 2021-2022.
- **Exercise:** Filter to see the sales for each year for only product 2.

Autofilling: Sequences

- Often we want to create a sequence in a column like 1, 2, 3, 4, etc
- If we have to do that for many values, it would be a lot of work.
- For common things, Excel can *autofill*.
- We do that as follows:
 - ① Type 1, 2, 3 in three neighboring vertical cells.
 - ② Select those cells.
 - ③ Click on the bottom-right of the selection and drag it down for as long as you want the sequence to go.
- This also works for other sequences. For example, 10, 20, 30.
- It also works for repeating the same number (like 1, 1, 1, 1).

Autofilling: Functions

- Often we want to get the sum of values for several columns. For example, for the 3 columns in:

	A	B	C
1	30	20	25
2	40	70	15

- If we have to type =SUM(A1:A2) many times (replacing A with B, C, etc.) it would be a lot of work.
- What we can do is (i) go to cell A3 and type the function =SUM(A1:A2), (ii) copy that cell, (iii) select cells B3 to C3, (iv) Paste (using Ctrl+V).
- This automatically changes the A to B and C in the formulas, giving the sums of the other columns (90 and 40).

Absolute Cell Referencing

- In the previous example, it could happen that we want to sum the values in column A using `=SUM(A1:A2)` and then paste **those** values on B3 and C3. That is, we want 70 repeated 3 times (instead of 70, 90 and 40).
- We can avoid the cell coordinates changing when we paste a function to different places using *absolute cell referencing*.
- We use a dollar symbol in front of the cell coordinates to do this.
- If we write `=SUM($A$1:$A$2)` in A3 and copy and paste that on B3 and C3, we will see 70 in each cell (instead of 70, 90 and 40).
- We can write `$A1` if we want the column part to stay fixed but allow the row part to vary.
- We can write `A$1` if we want the column part to vary but keep the row part fixed.

Pivot Tables

Suppose you are a greengrocer and you have data that look like this:

	A	B	C
1	Date	Category	Sales
2	01/10/2023	Fruit	3400
3	01/10/2023	Vegetables	2800
4	01/10/2023	Other	500
5	02/10/2023	Fruit	3600
6	02/10/2023	Vegetables	3000
7	02/10/2023	Other	600
8	03/10/2023	Fruit	3200
9	03/10/2023	Vegetables	1900
10	03/10/2023	Other	450
11	04/10/2023	Fruit	3500
12	04/10/2023	Vegetables	2200
13	04/10/2023	Other	550

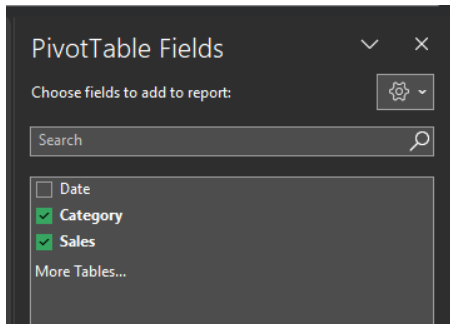
Suppose you want to summarize the data on this table.
For example:

- The total sales per product category
- The average daily sales per product category
- The total sales per day

We can do this using pivot tables.

Pivot Tables

- Select the table
- Go to Insert → PivotTable
- Choose either new or existing worksheet depending on where you want the new table.



- To get the total sales per category, tick the “Catorgy” and “Sales” columns.
- By default we will already get the desired table.
- Below we see the “Rows” are “Category” and the “Values” are “Sum of Sales”

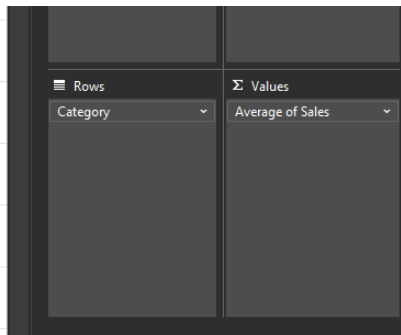
Row Labels	Sum of Sales
Fruit	13700
Other	2100
Vegetables	9900
Grand Total	25700

Pivot Tables

- To instead get the average daily sales per product category:
 - ▶ We click on “Sum of Sales” under Σ Values.
 - ▶ “Value Field Settings”
 - ▶ Click in “Average”.

The result should be:

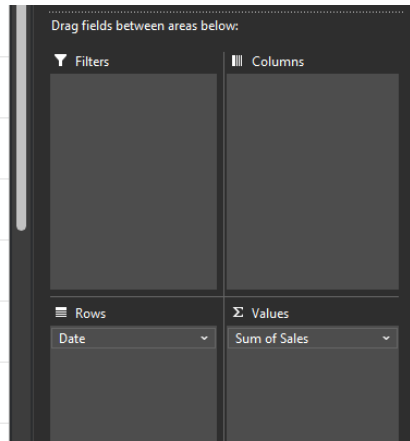
Row Labels	Average of Sales
Fruit	3425
Other	525
Vegetables	2475
Grand Total	2141.666667



Pivot Tables

- To get the total sales per day:
 - ▶ We change the columns to be “Date” and “Sales”.
 - ▶ We change the “Value Field Settings” under Σ Values back to “Sum”.

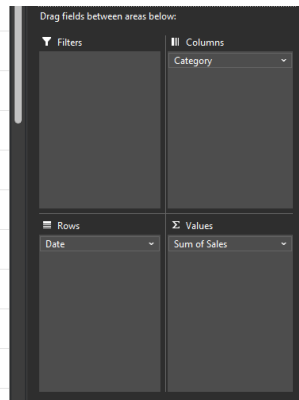
Row Labels	Sum of Sales
01/10/2023	6700
02/10/2023	7200
03/10/2023	5550
04/10/2023	6250
Grand Total	25700



Pivot Tables

- By ticking all of “Date”, “Category” and “Sales” we can obtain a more detailed table.
- By dragging “Category” to “Columns” and setting “Rows” to “Date”, we can get the sum of sales by date and category simultaneously:

Sum of Sales					
Column Labels					
Row Labels					
Fruit					
Other Vegetables					
Grand Total					
01/10/2023	3400	500	2800	6700	
02/10/2023	3600	600	3000	7200	
03/10/2023	3200	450	1900	5550	
04/10/2023	3500	550	2200	6250	
Grand Total	13700	2100	9900	25700	



VLOOKUP

Suppose you have a table with employee IDs and names, the like the table on the left:

	A	B	C	D	E	F	G	H	I
1	Raw Data Extract					Pay Report			
2									
3	Employee ID	Last Name	First Name			Employee ID	Pay	First Name	Last Name
4	110608	Doe	John			990678	\$84,289		
5	253072	Cline	Andy			830385	\$137,670		
6	352711	Smith	John			795574	\$190,024		
7	391006	Pan	Peter			580622	\$122,604		
8	392128	Favre	Bret			549457	\$111,709		
9	549457	Elway	John			392128	\$85,931		
10	580622	Manning	Eli			391006	\$168,114		
11	602693	Vick	Micheal			352711	\$89,627		
12	611810	Woods	Tiger			253072	\$149,946		
13	612235	Jordan	Micheal			612235	\$145,893		
14	795574	Stark	Tony			611810	\$64,757		
15	830385	Williams	Prince			602693	\$71,478		
16	990678	Pitt	Brad			110608	\$131,505		
17									

And you want to use this information to fill in the gaps in the table on the right.

VLOOKUP

- We can do this using the VLOOKUP function.
- More generally, we use VLOOKUP function to find a particular row of a table using a search query (with exact or approximate matching), and to return the value of a different column of that row.
- The VLOOKUP function takes 4 arguments:
 - ① The value you want to look up.
 - ② The range of values containing the table.
 - ③ The column number of the table we want to return for the match.
 - ④ Whether we want an approximate or exact match (TRUE for an approximate match, FALSE for an exact match).

VLOOKUP

In our example:

- ❶ The value we want to look up: We use the employee ID, so F4 for the first employee.
- ❷ The range of values containing the table: This is \$A\$4:\$C\$16, where we use absolute cell referencing as we want to copy the function for the remaining employees, but the table stays fixed.
- ❸ The column number of the table we want to return for the match: This is 3 for the “First Name” and 2 for the “Last Name”
- ❹ Whether we want an approximate or exact match: For employee IDs we should use exact matching, so FALSE.

So: =VLOOKUP(F4; \$A\$4:\$C\$16; 3; FALSE) for “First Name”